

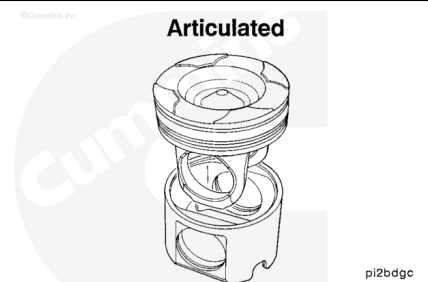
Trainings ▾

General Information

All M Series engines are equipped with articulated pistons.

The articulated piston is a two-piece piston consisting of a forged-steel crown and an aluminum skirt.

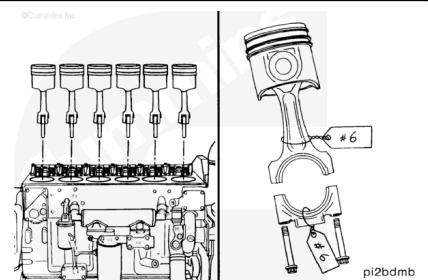
A closed chamber oil gallery is located on the underside of the crown of the piston to provide more oil contact with the piston and give better piston cooling. Piston cooling nozzle targeting is critical because of the closed chamber oil gallery.



Remove

The piston and connecting rod **must** be removed as an assembly.

- Remove the piston and connecting rod. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/101/101-001-054-tr.html>)



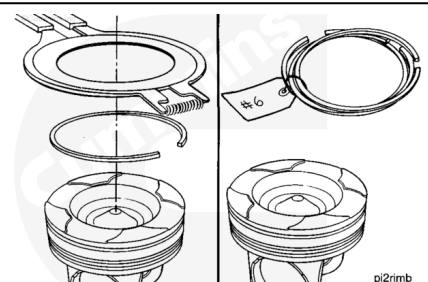
Disassemble

Articulated Piston

Disassemble the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1.

(</qs3/pubsys2/xml/en/procedures/101/101-001-054-tr.html>)

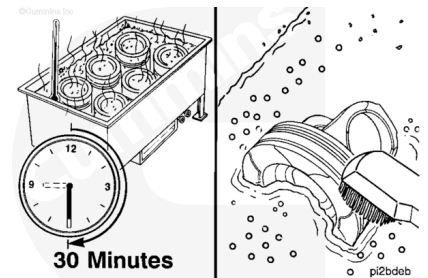
Inspect the piston rings. Refer to Procedure 001-047 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-047-tr.html>)



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ CAUTION ⚠

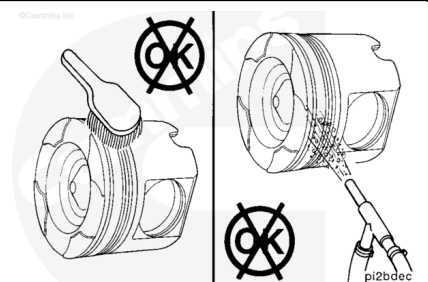
Be sure the cleaning solvent is approved for aluminum. Damage to the pistons can result if the wrong solution is used.

Allow the pistons to soak for a minimum of 30 minutes in a tank containing an approved cleaning solvent for aluminum.

Use a hot, soapy solution and a nonmetallic brush to remove carbon deposits.

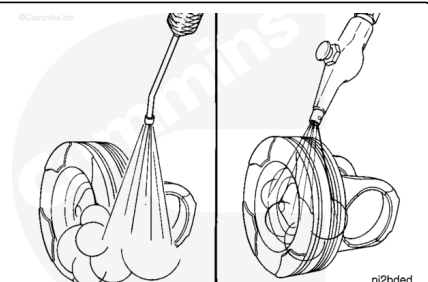
⚠ CAUTION ⚠

Do not use a metal brush. A metal brush will damage the piston ring grooves. Do not use glass beads to clean the grooves. Walnut shell or plastic bead, Part Number 3822735, blasting can be used on ring grooves on the dome or crown of the piston. Use the minimum effective pressure, and do not concentrate the spray in one area for an extended period of time. The recommended blast pressure for plastic bead blasting is 276 kPa [40 psi]. Do not use glass beads or walnut shell blasting on the aluminum piston pin bores or articulated piston skirts. This can cause piston pin bore damage.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

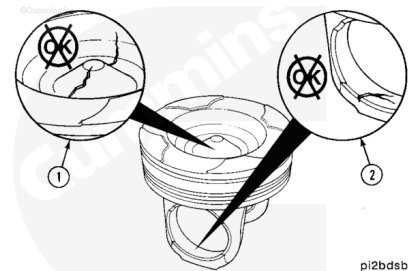
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use steam to clean the pistons.

Dry with compressed air.

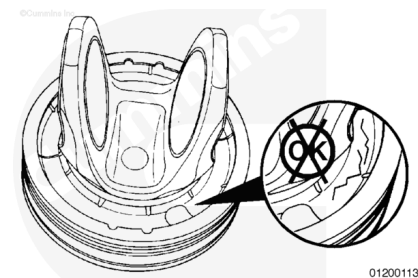
Inspect the piston bowl (1) and crown piston pin bore (2) for cracks or damage.

Do **not** use pistons with cracks. If the crown is cracked, it **must** be replaced.



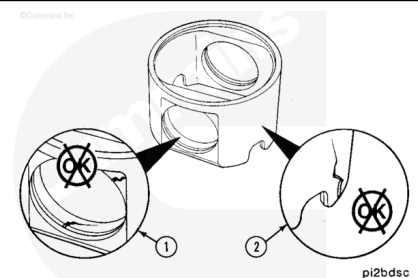
Inspect the piston gallery cover plate in the piston crown for cracks or looseness.

If the gallery is cracked or loose, the piston crown **must** be replaced.



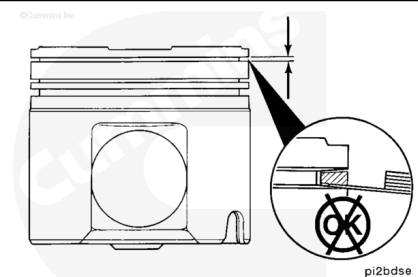
Inspect the piston skirt pin bore (1) and sides (2) for cracks or damage.

Do **not** use pistons with cracks. If the skirt is cracked, it **must** be replaced.



The ring groove can be inspected with a new ring and a feeler gauge.

Hold a new ring in the groove even with the outside diameter of the piston. Install a 0.15-mm [0.006-in] feeler gauge. If the feeler gauge enters the groove without resistance, there is too much wear. Replace the piston.



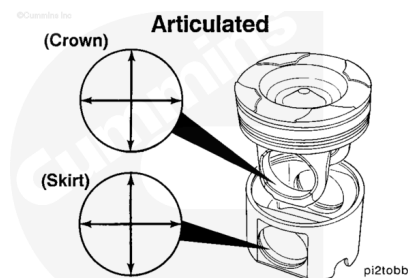
Measure the piston pin bore inside diameter.

Articulated Piston Pin Bore Inside Diameter (Crown)

mm		in
54.040	MIN	2.1276
54.055	MAX	2.1281

Articulated Piston Pin Bore Inside Diameter (Skirt)

mm		in
54.007	MIN	2.1263
54.015	MAX	2.1266



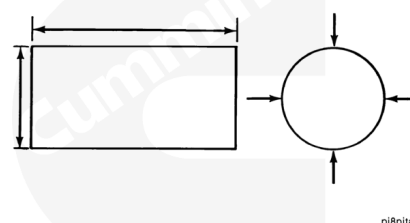
Measure the piston pin outside diameter and length.

Piston Pin Outside Diameter

mm		in
53.997	MIN	2.1259
54.003	MAX	2.1261

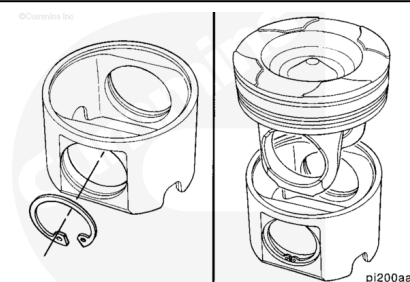
Piston Pin Length

mm		in
101.70	MIN	4.0039
102.00	MAX	4.0157



Assemble

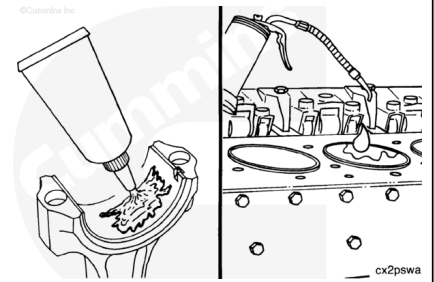
- Assemble the piston and connecting rod. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/101/101-001-054-tr.html>)



Install

The piston and connecting rod **must** be installed as an assembly.

- Install the piston and connecting rod assembly. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/101/101-001-054-tr.html)



If the malfunction resulted in coolant, oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system **must** be inspected. Refer to Procedure 014-013 in Section 14.
(/qs3/pubsys2/xml/en/procedures/101/101-014-013.html)